Kakumanu Jaswanth

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.89
Sri Chaitanya Junior College	Class XII (Board of Intermediate Education Telangana)	2023	99.00%
Triveni Talent School	Class X (Board of Secondary Education Telangana)	2021	100.00%

WORK EXPERIENCE

ICICI Bank

Software Engineer Intern

- **Built a C++17 FX pricing simulator** with synthetic Retail, HNI and Corporate customers across **5 INR currency pairs**, modelling trade size, loyalty, market volatility and one-sided daily flow. It replays identical customer flow against each strategy, so results compare fairly, and it is benchmarked against a per-segment tuned static rate sheet fitted on separate history, across **10 random seeds** of **60 trading days and 500 quotes** each.
- **Implemented a from-scratch logistic regression** (Newton's method) that predicts the probability that a customer accepts a quote at a given spread. The pricing engine tries candidate spreads and quotes the one that maximises **P(accept | spread) × revenue**. Simulated profit rose **5.2% overall (7.8% after warm-up)** over the tuned baseline.
- **Added inventory-aware pricing** that skews quotes to offset the bank's open currency positions. It cut hedging cost by **28%** and daily price risk by **18%** with no loss of profit. I also built an **HTML/JS dashboard** to visualise the results.

PROJECTS

Stack- Based Scripting Language & Virtual Machine | Git-hub

- Built a **stack-based VM/compiler in C++** with a lexer, recursive-descent parser, and **AST-to-bytecode compilation**, featuring a custom **24-opcode instruction set** and compile-time variable resolution.
- Implemented **stack-based bytecode execution**, backpatched relative jumps for if/else/while control flow, and panic-mode error recovery; validated functionality with **137 tests.**
- Benchmarked **switch vs. computed-goto dispatch** over **92M retired instructions,** achieving **1.69 ns/instruction** compared to **2.29 ns/instruction**.

LOB-engine: Limit Order Book & Matching Engine | Git-hub

- Built a single-symbol exchange matching engine in **C++17** using price-time priority, supporting limit and market orders plus order cancellation and modification.
- Verified correctness with **99 GoogleTest cases**, including randomized testing against an independent reference implementation and planted-bug checks to confirm the tests catch errors.
- Benchmarked a tree-based order book against a cache-friendly flat-array design on **20M** synthetic orders. The flat array was **1.35×** faster (**6.0M orders/sec, 229 ns median latency**), with tail latency **(p99/p99.9**) measured using calibrated CPU cycle counters.

Optimal Route Finder Between Indian Cities | Git-hub

- Developed a **C++ route optimization system** using **Dijkstra's algorithm** to compute shortest railway routes between Indian cities using real-world Indian Railways data.
- Modeled the railway network as a **weighted graph,** incorporating intercity distances and station connectivity to determine optimal travel paths across major metropolitan and significant cities.
- Implemented **efficient graph traversal and shortest-path computation** to identify minimum-distance routes across interconnected railway stations.

SKILLS

- **Programming Languages:** C, C++, Python(Familiar)
- **Technologies & Frameworks:** Web(HTML, CSS, ReactJs)
- **Database:** MySQL
- **Misc:** Git, GitHub, SolidWorks
- **Problem Solving:** Data Structures and Algorithms, Object Oriented Programming

ACHIEVEMENTS

- **Codeforces:** Ranked 905 globally in Round 1121(Div. 2) among 17292 participants; achieved a max rating 1592
- **LeetCode**: Ranked **519** globally among **45,329 participants** in Weekly Contest 518; achieved a max rating 1550
- **Competitive Programming:** Solved **1000+ problems** across platforms like Codeforces, LeetCode, HackerRank and CSES.

RELEVANT COURSES

Mathematics: Calculus 1(A-), Calculus 2(A), Linear Algebra(A-), Differential Equations(A-), Probability(A-), Transform Techniques(A-), Statistics(A-)

POSITIONS OF RESPONSIBILITY

Transport Head, Student Gymkhana

- Managed and coordinated college bus operations throughout the tenure, overseeing transportation schedules, route planning, and smooth bus movements to ensure efficient and reliable student transport

Core Member, Aero Club

- Contributed to the design, assembly, and testing of RC aircraft and drones throughout the tenure. Collaborated with team members on aerospace projects, supporting hands-on aircraft development and drone engineering activities.